

HR Analytics Dashboard

IBM Employee Attrition Analysis Report

Dataset	Employees	Tools	Year
IBM HR Analytics	1,470 Records	Python MySQL Tableau	2026

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Executive Summary

This report presents a comprehensive analysis of employee attrition at IBM, based on HR data covering 1,470 employees across multiple departments. The objective was to identify the key factors contributing to employee turnover and provide data-driven recommendations to support HR decision-making.

Metric	Value	Insight
Total Employees	1,470	Full workforce analyzed
Employees Left	237	Confirmed attrition cases
Attrition Rate	16.12%	1 in 6 employees leaving
Avg Monthly Income	\$6,503	Overall workforce average
Avg Employee Age	36.9 years	Mid-career workforce
Avg Experience	11.3 years	Total working years

Key Findings Summary

- 16.12% overall attrition rate — significantly above healthy industry benchmark of 10%
- Sales department is the highest risk area with 20.63% attrition rate
- Young employees (18-25) are leaving at an alarming rate of 34.78%
- Overtime is a major driver — workers doing overtime leave at 3x the rate of others
- Low salary correlates strongly with attrition — leavers earned ~\$2,000 less per month
- Poor work-life balance (score 1) is associated with 31.25% attrition rate

Interactive Dashboard

The following dashboard was built in Tableau Public and provides an interactive view of all attrition metrics. It includes 7 charts covering department, age group, job role, salary, overtime, job satisfaction, and work-life balance analysis.

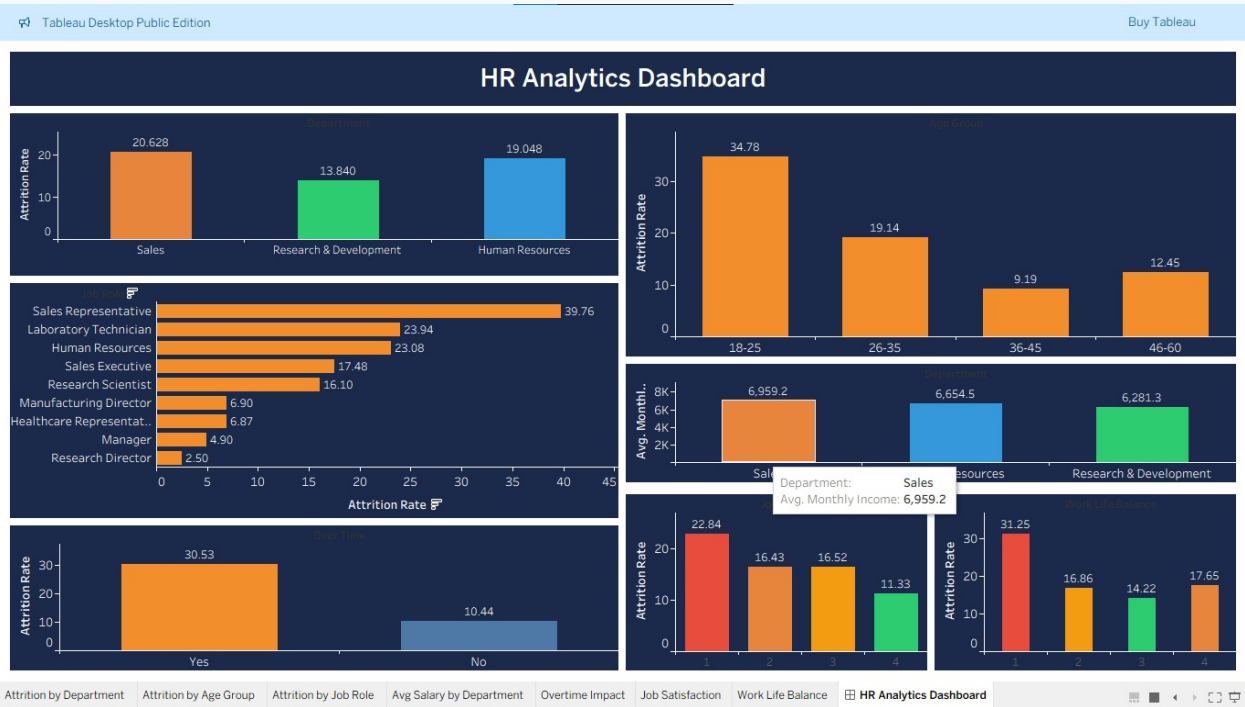


Figure 1: HR Analytics Dashboard — Tableau Public (Interactive)

Dashboard Feature	Details
Total Charts	7 interactive charts
Chart Types	Bar, Horizontal Bar, Column
Filters	Department, Job Role, Overtime slicers
Color Theme	Dark Navy (#1B2A4A) with Orange (#E8853D) accents
Platform	Tableau Public 2025.2
Live Link	public.tableau.com/app/profile/mohd.badruzama.ansari

Department & Age Group Analysis

Attrition by Department

The Sales department records the highest attrition rate at 20.63%, followed by Human Resources at 19.05%. Research & Development, despite being the largest department, has a comparatively lower attrition rate of 13.84%.

Department	Total Employees	Employees Left	Attrition Rate	Avg Salary
Sales	446	92	20.63%	\$6,959
Human Resources	63	12	19.05%	\$6,655
Research & Development	961	133	13.84%	\$6,281

Attrition by Age Group

Young employees between 18-25 years show the highest attrition at 34.78%, nearly double that of the 26-35 age group. This suggests challenges in retaining early-career talent, possibly due to career growth expectations or compensation gaps.

Age Group	Attrition Rate	Risk Level	Likely Reason
18 - 25	34.78%	CRITICAL	Career growth & salary gaps
26 - 35	19.14%	HIGH	Better opportunities elsewhere
46 - 60	12.45%	MEDIUM	Early retirement considerations
36 - 45	9.19%	LOW	Stable career stage

Job Role & Salary Analysis

Attrition by Job Role

Sales Representatives have an alarmingly high attrition rate of 39.76% — nearly 1 in 2 employees in this role leaves. Laboratory Technicians and Human Resources roles also show above-average attrition.

Job Role	Attrition Rate	Risk Level
Sales Representative	39.76%	CRITICAL
Laboratory Technician	23.94%	HIGH
Human Resources	23.08%	HIGH
Sales Executive	17.48%	MEDIUM
Research Scientist	16.10%	MEDIUM
Manufacturing Director	6.90%	LOW
Healthcare Representative	6.87%	LOW
Manager	4.90%	LOW
Research Director	2.50%	LOW

Salary vs Attrition

A clear salary gap exists between employees who stayed and those who left. Employees who left the company earned approximately \$2,000 less per month on average, strongly suggesting that compensation is a primary driver of attrition.

Group	Avg Monthly Income	Difference
Employees who STAYED	\$6,833	--
Employees who LEFT	\$4,787	-\$2,046 less

Risk Factors Analysis

Overtime Impact

Overtime is one of the strongest predictors of attrition. Employees working overtime have a 30.53% attrition rate compared to just 10.44% for those who do not — a 3x difference.

Overtime Status	Total Employees	Attrition Rate	Risk
Yes (Working Overtime)	416	30.53%	CRITICAL
No (No Overtime)	1,054	10.44%	NORMAL

Job Satisfaction & Work-Life Balance

Both job satisfaction and work-life balance show a clear inverse relationship with attrition. Employees with the lowest scores are significantly more likely to leave.

Factor	Score 1 (Worst)	Score 2	Score 3	Score 4 (Best)
Job Satisfaction	22.84%	16.43%	16.52%	11.33%
Work-Life Balance	31.25%	16.86%	14.22%	17.65%

Business Recommendations

Based on the data analysis, the following actionable recommendations are proposed to reduce employee attrition:

1. Review Sales Compensation Packages

Sales Representatives have the highest attrition at 39.76%. A salary review and performance-based incentive restructuring could significantly reduce turnover in this critical revenue-generating role.

2. Reduce Mandatory Overtime

Overtime workers leave at 3x the rate of others. Implementing better workload distribution, hiring additional staff for high-demand periods, and monitoring overtime hours can directly reduce this risk.

3. Develop Early Career Retention Programs

Employees aged 18-25 have a 34.78% attrition rate. Structured mentorship programs, clear career progression paths, and competitive entry-level salaries can help retain young talent.

4. Conduct Regular Salary Benchmarking

Leavers earned ~\$2,000 less per month than stayers. Annual salary benchmarking against industry standards and merit-based raises can address compensation-driven attrition.

5. Improve Work-Life Balance Initiatives

Score 1 employees show 31.25% attrition. Flexible working hours, remote work options, and wellness programs can improve work-life balance scores and reduce turnover.

6. Focus Retention Efforts on Sales & HR Departments

Both departments show ~20% attrition. Department-specific exit interviews and engagement surveys can help identify and address root causes more precisely.

Conclusion

This analysis of IBM's HR dataset reveals that employee attrition is driven by a combination of compensation gaps, excessive overtime, poor work-life balance, and limited career growth opportunities — particularly affecting younger employees and those in Sales and HR roles.

The 16.12% overall attrition rate represents a significant business risk. Replacing an employee typically costs 50-200% of their annual salary in recruitment, onboarding, and lost productivity. For an organization of 1,470 employees, reducing attrition by even 5 percentage points could result in substantial cost savings.

The interactive Tableau dashboard provides HR teams with a real-time view of attrition patterns, enabling proactive intervention before at-risk employees decide to leave.

Project Links

Resource	Link
Live Dashboard	https://public.tableau.com/app/profile/mohd.badruzama.ansari/viz/HR-Analytics-Dashboard_17804982697270/HRAnalyticsDashboard
GitHub Repository	https://github.com/ansaribadruzama/HR-Analytics-Dashboard
Dataset Source	https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

Author Information

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Target Role	Data Analyst (Fresher)
Skills	Python MySQL Tableau Power BI Excel